

**Master of Computer Application (M.C.A.) (Two Year Programme) –
2nd Semester (CBEGS) (Batch 2024-26)
(2525)**

Paper : CSL-4094 Open Source Software

Time allowed: 3 hrs.

Max. Marks: 70

Note: Candidates are required to attempt five questions, selecting at least one question from each section. The fifth question may be attempted from any section. All questions carry equal marks.

Section-A

- ✓ 1. a) Why do companies choose FLOSS business models over proprietary software? (7)
- b) How does the Onion Model Illustrate the hierarchy of contributor roles in FLOSS projects, and are these roles well-defined? (7)
2. a) What factors do businesses evaluate before contributing code to a FLOSS project used internally? (6)
- b) How does a wrong choice between permissive (MIT) and copyleft (GPL) licenses lead to legal issues for a company? (8)

Section-B

3. a) How does Git and GitHub interact, and what kind of errors do developers usually encounter while working on a project? (7)
- b) What key steps would you follow to develop a software project of your choice, from initial idea to a functional version? (7)
4. a) How does Git's branching and merging prevent teams from overwriting work in a FLOSS project? Explain with a brief example. (8)
- b) How do humanitarian FLOSS projects contribute to solving societal problems in various ways? (6)

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(2)

Section-C

- ✓ 5. a) What documentation does play a significant role in the overall success and sustainability of a FLOSS project? (8)
- b) What quality assessment models can developers utilize to evaluate FLOSS tools before incorporating them into their projects? (6)
6. a) What types of potential security risks and vulnerabilities are commonly found in FLOSS projects? Cite at least one relevant security report to support your answer. (8)
- b) Why do some FLOSS projects lag in updates, and how does this impact the overall project success? (6)

Section-D

- ✓ 7. a) How does a commercial firm's involvement affect the overall sustainability of a FLOSS project? (7)
- b) How do different entities within the Open Source Software ecosystem interact and depend on each other? (7)
8. a) Explain the Open Source Software (OSS) development process and how different models exhibit varying levels of openness and collaboration. (8)
- b) How can RESTful APIs and endpoints be used to mine data from software repositories, and what are some common challenges encountered in this process? (6)
